Yist Tingyang YU

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RESEARCH INTERESTS

Developing generative and geometric deep learning frameworks for tissue modeling and single-cell spatial analysis. Committed to extracting critical biological insights from the intersection of spatial omics and multi-omics.

EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL) Sep. 2023 – Present
Ph.D. in Communication and Computer Sciences (EDIC) Lausanne, Switzerland

- Advised by Prof. Maria Brbic at the Machine Learning for Biomedicine (MLBio) Lab.
- EDIC Computer Science Fellowship Recipient (Direct Doctorate).

The Chinese University of Hong Kong (CUHK) Aug. 2019 – July 2023
B.Sc. in Mathematics and Information Engineering (MIEG) With Honours, First Class Hong Kong SAR, China

- Ranked among the **top 3** students among MIEG students admitted in 2019-20. (Exact rank is unavailable for students)
- **Winter semester exchange program in Faculty of Applied Science & Engineering, University of Toronto (UoT), 2021-22.**
- Ranked **top 0.4% among 163,000 science students** in the National College Entrance Exam.

PUBLICATIONS AND PREPRINTS

1. **Tissue reassembly with generative AI** [paper] [project]
   Tingyang Yu, Chanakya Ekbote, Nikita Morozov, Antonio Herrera, Jiashuo Fan, Albert Dominguez Mantes, Salvatore Novello, Hilal Lashuel, Pascal Frossard, Stéphane d'Ascoli, Gioele La Manno, and Maria Brbić.
   Nature Methods in Press
   ICML 2025 GenBio Workshop
2. **SCMBench: Benchmarking Domain-specific and Foundation Models for Single-cell Multi-omics Data Integration** [paper]
   Yixuan Wang, Yimin Fan, Xuesong Wang, Tingyang Yu, Yongshuo Zong, Xinyuan Liu, Meitong Liu, Qing Li, Kin hei Lee, Khachatur Dallakyan, Gengjie Jia, Jiao Yuan, Ting-Fung Chan, Xin Gao, Irwin King, and Yu Li.
   Nature Communication
3. **Fool Your (Vision and) Language Model With Embarrassingly Simple Permutations.** [paper]
   Yongshuo Zong, Tingyang Yu, Bingchen Zhao, Ruchika Chavhan, and Timothy Hospedales.
   ICML 2024
4. **Safety Fine-Tuning at (Almost) No Cost: A Baseline for Vision Large Language Models.** [paper]
   Yongshuo Zong, Ondrej Bohdal, Tingyang Yu, Yongxin Yang, and Timothy Hospedales.
   ICML 2024
5. **Contrastive Cycle Adversarial Autoencoders for Single-cell Multiomics Alignment and Integration** [paper]
   Xuesong Wang, Zhihang Hu, Tingyang YU, Ruijie Wang, Yumeng Wei, Juan Shu, Jianzhu Ma, Yu Li
   Bioinformatics 2023
6. **scMinerva: an Unsupervised Graph Learning Framework with Label-efficient Fine-tuning for Single-cell Multi-omics Integrative Analysis** [paper]
   Tingyang Yu, Yongshuo Zong, Yixuan Wang, Xuesong Wang, Yu Li.
   biorXiv Preprint 2022
7. **conST: an interpretable multi-modal contrastive learning framework for spatial transcriptomics** [paper]
   Yongshuo Zong, Tingyang YU, Xuesong Wang, Yixuan Wang, Zhihang Hu, Yu Li
   biorXiv Preprint 2021

RESEARCH EXPERIENCE

Subgraph Augmentation for Graph Neural Networks Sep. 2022 – May, 2023
Final year project supervised by Professor Irwin King (CUHK) Hong Kong SAR, China

- Conducted thorough experiments which showed that this strategy outperformed state-of-the-art sampling strategies published at ICLR 2022 on graph classification by 3% on TU datasets. It further reduced the space complexity up to 55% on dataset IMDB-MULTI and 10-20% training time on dataset NCI1 compared to the second best strategy.

Projection Robust Unbalanced Optimal Transport Feb. 2022 – Dec. 2022
Research Assistant to Professor Ma Shiqian (University of California, Davis) California, USA

- Presented a new model called projection robust unbalanced optimal transport (PRUOT) to alleviate the curse of dimensionality for computing the Wasserstein distance between unbalanced distributions.
- Illustrated a Riemannian optimization algorithm RGAS-UOT for solving this new model and proved its non-trivial finite-time convergence guarantee to obtain the stationary point. Numerical experiments were given to demonstrate the advantages of the PRUOT model in high-dimensional cases. **The work was presented in this report**.

Single-cell Multi-omics Integration with Random Walk and Graph Neural Network April, 2021 – Sep., 2023
Research Assistant to Professor Li Yu (CUHK) Hong Kong SAR, China

- Presented *scMinerva*, an unsupervised Graph Convolutional Network framework for single-cell multi-omics integration, which featured a novel algorithm *omics2vec* that enabled random walk algorithm for heterogeneous graphs.
- scMinerva achieved superior classification performance among 7 state-of-the-art methods on 6 real-world datasets and improved classification accuracy up to 20% on dataset GSE128639 when classifying to 27 classes.
- Won the Best Project Award for Summer Research Internship of CUHK, 2020-21 **(Top 10% out of 46 projects)**.

ACADEMIC SERVICES

Selected Posters and Invited Talks

- ***Poster***: Generating Tissue Structure from Gene Expression, **Single Cell Genomics Conference**, Greece, 2024
- ***Poster***: Generating Tissue Structure from Gene Expression, **Human Cell Atlas General Meeting**, Milan, 2024
- ***Invited Talk***: Tissue Reassembly Using Generative AI, **Mila Multiomics Reading Group**, Online, August 6, 2025
- ***Invited Talk***: Tissue Reassembly Using Generative AI, **Harvard Medical School**, Boston, December 6, 2025

Organizer and Program Chair

- Student Organizer and Program Chair, **NeurIPS 2024 AIDurgX Workshop**
- Organizer and Program Chair, **ICML 2025 GenBio Workshop**

Reviewer

- **RECOMB 2024** (1 paper)
- **ICLR 2024** MLGenX Workshop (2 papers)
- **NeurIPS 2024** AIDurgX Workshop (6 papers)
- **ICLR 2025** MLGenX Workshop (2 papers)

Student Supervision

- Nikita Morozov (HSE): **Spatial Reconstruction in 3D from Gene Expression**, Summer 2024
- Ananya Gupta (EPFL): **Integrating Morphology with Gene Expression for Tissue Reassembly**, Spring 2025
- Manon Benarafa (EPFL): **Frequent Pattern Mining for Graphs and Point Clouds**, Autumn 2025
- Saba Nasiri (EPFL): **Scalable Motif Discovery using Spatial Data**, Spring 2026

AWARDS AND SCHOLARSHIPS

- EDIC Computer Science Fellowship (54,000 CHF), 2023–24 — *EDIC, EPFL*
- Academic Excellence Scholarship for Non-local Students (30,000 HKD), 2022–23 — *Faculty of Science, CUHK*
- Professor Charles K. Kao Exchange Scholarship (50,000 HKD), 2021–22 — *The Charles K. Kao Foundation*
- Dean's List (Top 10%) for 2019–20, 2020–21, 2022–23 — *CUHK*
- ELITE Stream Scholarship (20,000 HKD), 2019–20, 2020–21 — *ELITE Stream*
- Best Project Award for Summer Research Internship, 2020–21 — *Faculty of Engineering, CUHK*
- Second Prize, National Physics Competition for Senior High, 2018–19 — *National Physics Competition Committee*
- Second Prize, National Mathematics Competition for Senior High, 2018–19 — *Chinese Mathematics Society*
- Best Debater Award (Hong Kong Region), 2020 Hangzhou Mandarin Debate Grand Prix (256 teams), 2020 — *Virtual*
- Champion, 1st FENGYE Cup Mandarin Debate Competition (32 teams), 2020 — *Virtual*
- Best Debater Award, 13th Xinwei–Huaxia Mandarin Debate World Cup, 2020 — *University of New South Wales*

LEADERSHIP AND COMMUNITY CONTRIBUTIONS

Mandarin Debate Team of CUHK Sep. 2019 – May 2021
Captain Hong Kong SAR, China

- Prepared over 9 hours of weekly debate training materials for 120+ team members and managed logistics for team registration and participation in 10+ international competitions.
- Founded the team's PR group to publicize competition results and coordinate collaborations. Secured over 30,000 CNY in sponsorships from corporate partners to support international competitions and debate exchange visits.

Hong Kong Dog Rescue Jan. 2023 – Mar. 2023
Part-time Volunteer Hong Kong SAR, China

- Maintained dog kennels, supported dogs with health issues, and facilitated socialization among over 600 dogs at the Tai Po Homing Centre. Assisted in hosting Homing Centre visits and promoting HKDR's mission to improve public understanding of dog welfare and reduce pet abandonment.

Women+ in IC at EPFL Sep. 2023 – Present
President and Committee Member Lausanne, Switzerland

- Organize seminars and workshops that provide platforms for female researchers and engineers within and beyond the Communication and Computer Sciences departments to share experiences, build connections, and inspire others. Built a local community of over 200 members, fostering a safe and inclusive environment for all.
- Led the design of the organization's first official logo, symbolizing themes of gender equality and social justice.

SKILLS & MISC.

- *Language* English (professional), Mandarin (native), Cantonese (fluent).
- *Programming* Matlab, C, R, Python {Pytorch, Tensorflow, NetworkX, Pandas, Numpy, sklearn}.
- *Tools* LaTeX, Linux, Anaconda, Git, MongoDB, Docker, WanDB, Abode Illustrator, Slurm.
- *Hobbies* Writing, swimming, running, certified open water diver (PADI), *etc.*